

Smart-Chatbot

AROL Group

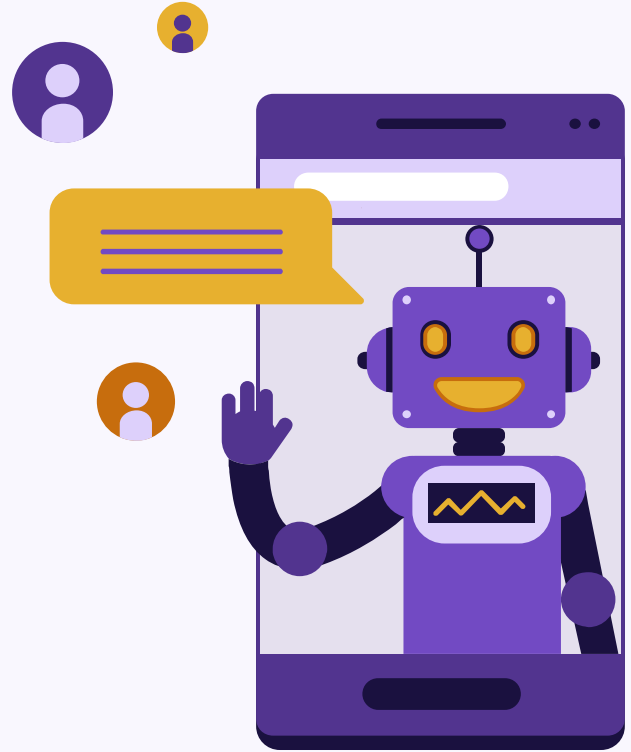


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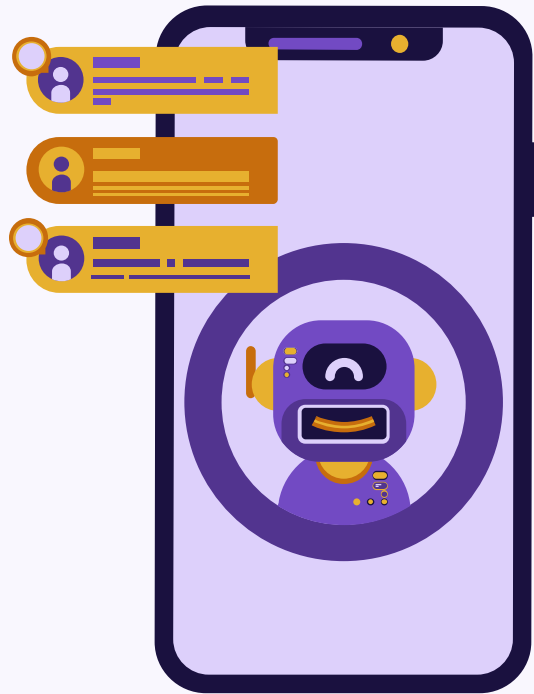
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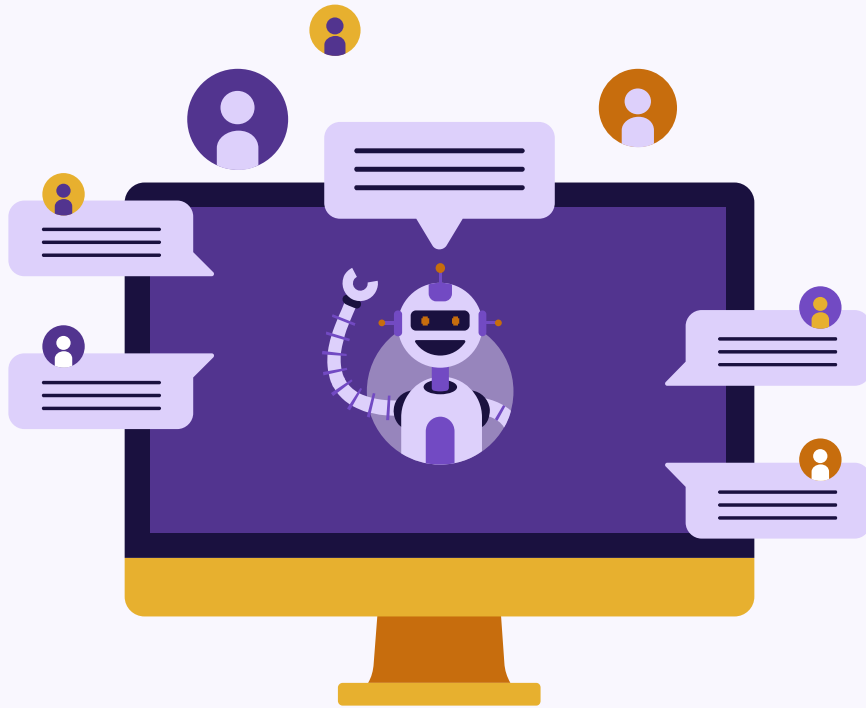
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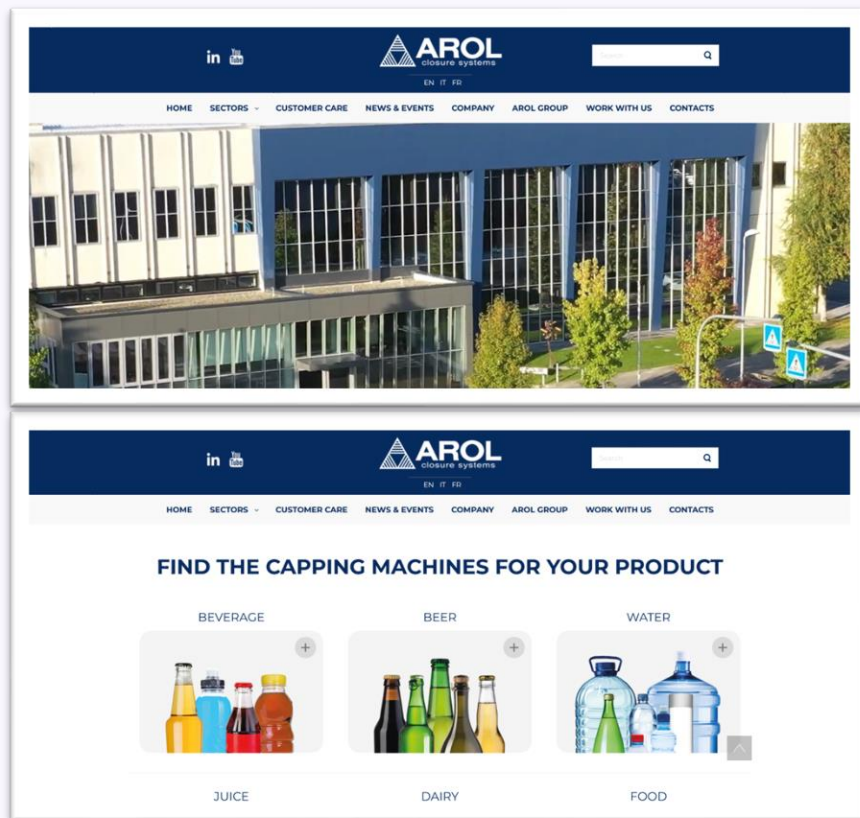
01

Web Crawling and Data Scraping

Web Crawling and Data Scraping

Scrapy Framework

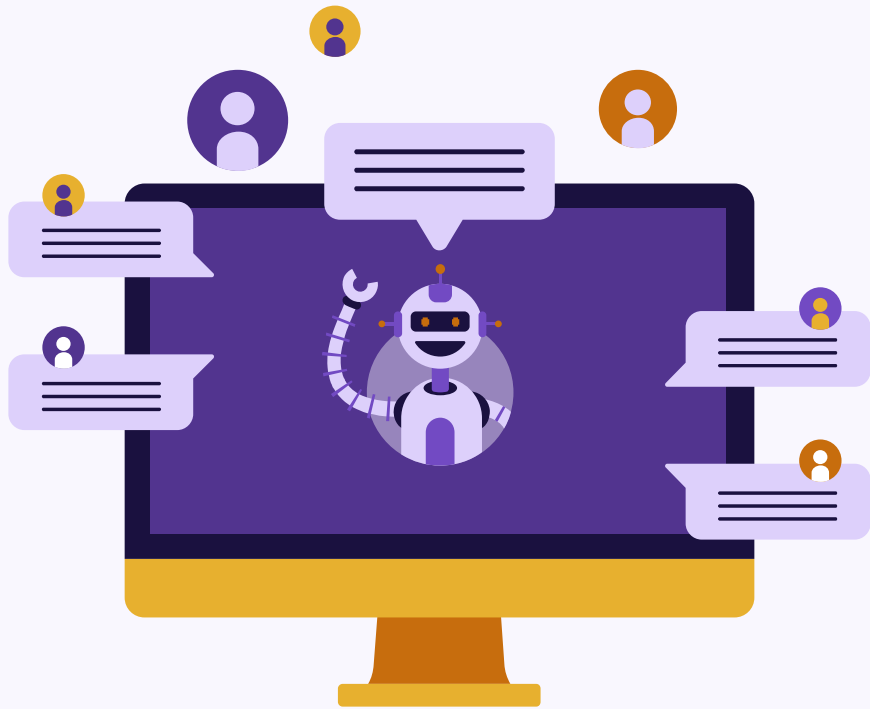
- **Web crawler:** *Scrapy framework*
- **Task:** Data scraping from AROL Group websites
- **Goal:** Filtered, structured data for LLM training and validation
- **Steps:**
 1. Collect structured web content
 2. Clean HTML pages into usable formats
 3. Generate analyzable output for model building



Web Crawling and Data Scraping

Parameters Configuration

```
custom_settings = {  
    "DEPTH_LIMIT": 3,  
    "ROBOTSTXT_OBEY": True,  
    "USER_AGENT": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36",  
    "DOWNLOAD_DELAY": 2,  
    "CONCURRENT_REQUESTS_PER_DOMAIN": 8,  
    "LOG_LEVEL": "INFO",  
    "FEEDS": {  
        "output.jsonl": {  
            "format": "jsonlines",  
            "encoding": "utf8",  
            "overwrite": True  
        }  
    }  
}
```



02

Data Preprocessing

Data Preprocessing

Question/Answer Format

- **Output file:** Contains data for training and validation
- **Issue:** Includes redundant info (scripts, styles, metadata)
- **Solution:** Process file to extract relevant content
- **Result:** Data organized in QA *format* for:
 - Supervised training
 - Targeted inference

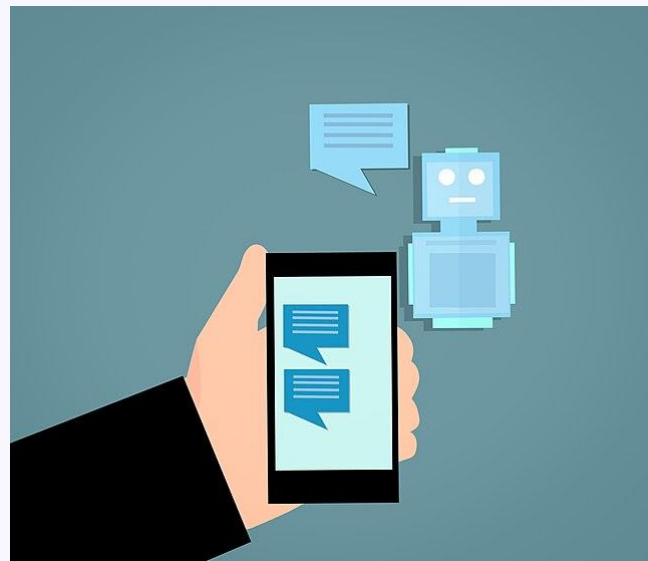


Image source: "Piqsels.com-id-zbxec.jpg," available at Piqsels, licensed under CC0/Public Domain Dedication.

```
{  
  "question": "Where can I find AROL's legal notes?",  
  "answer": "Our legal notes are available in the 'Terms & Conditions' section of our website."  
}
```

Data Preprocessing

Prompt Template

```
data_prompt = """
You are a customer support assistant for AROL Group, specialized in bottle caps and capping technologies.
Your goal is to provide accurate, clear, and helpful responses about AROL Group's products and processes.

### question:
{}

--- Instructions ---
- Provide a concise and informative response about bottle cap manufacturing or capping technology.
- If technical details or product features are mentioned, explain them simply.
- If concerns are raised, offer relevant recommendations or solutions.
- Keep the answer focused on the specific query.

### answer:
{}

""".strip() # Using strip to avoid trailing newlines at the end
```

Data Preprocessing

Hugging Face Datasets

- **Step:** Convert QA pairs into *Hugging Face Datasets*
- **Purpose:** Optimize for batching, tokenization, and parallel processing
- **Result:** Suitable for ML tasks like training and validation

```
training_data = Dataset.from_pandas(df_training)
training_data = training_data.map(formatting_prompt, batched=True)

validation_data = Dataset.from_pandas(df_validation)
validation_data = validation_data.map(formatting_prompt, batched=True)
```

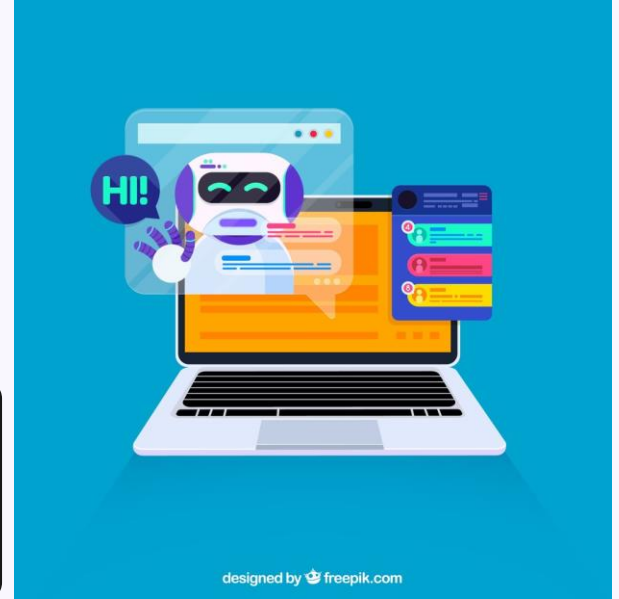
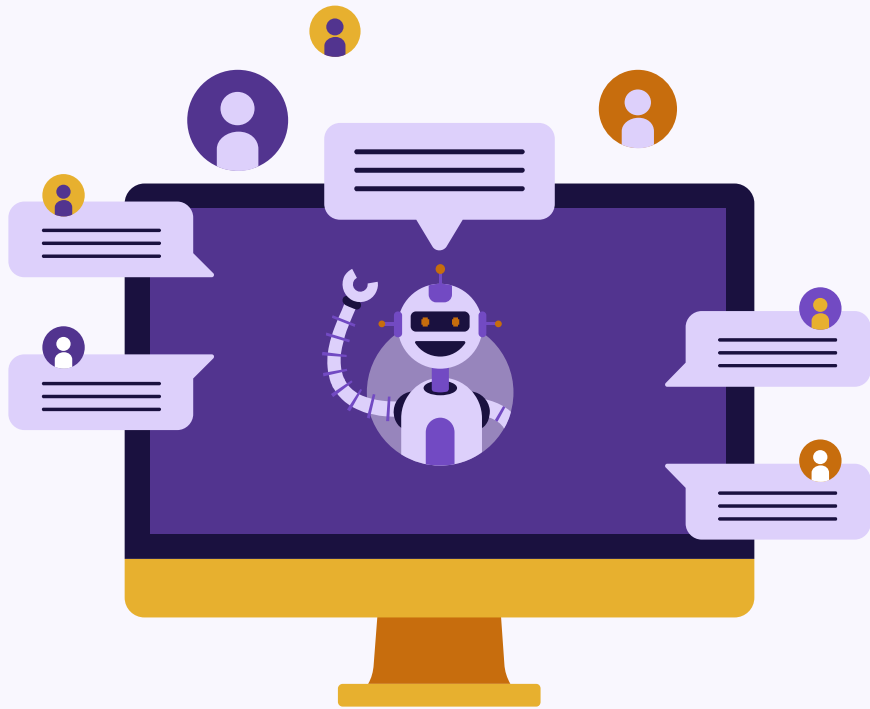


Image source: "Chatbot concept background with mobile device," designed by Freepik. Attribution required as per the license. Retrieved from Freepik (<https://www.freepik.com>).



03

Model Fine-Tuning

Model Fine-Tuning

Setting up the Pretrained Model

- **Model setup:** Started with *Llama 3.2* (1B parameters)
- **Issue:** Poor chatbot performance.
- **Solution:** Switched to 3B parameters, achieving better results
- **First fine-tuning attempt:** Used *4-bit quantization*
- **Problem:** Significant data loss, inefficient fine-tuning
- **Final solution:** *Full precision* (no quantization) for simplicity and effectiveness

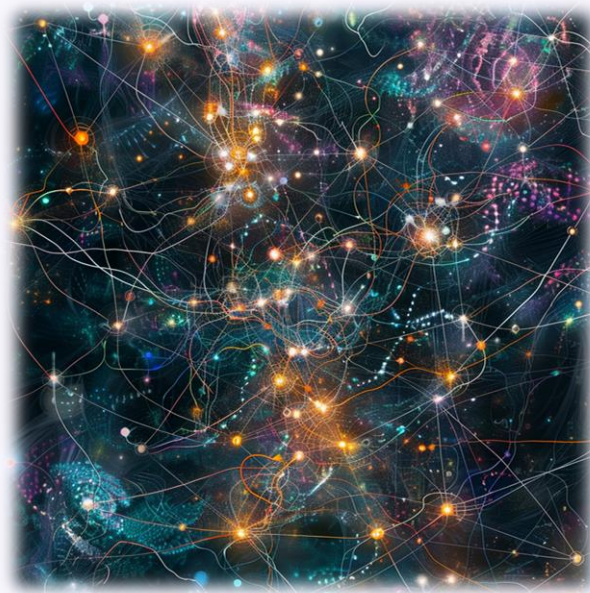


Image source: "Neural network - Midjourney and Grok.png," created using Midjourney with a prompt suggested by Grok. Available in the public domain.

Model Fine-Tuning

Low-Rank Adaptation

- **Method:** *LoRa* (Low-Rank Adaptation) via *PEFT*
- **Applied to:** Pretrained model
- **Key feature:** Trains specific model parts, not the entire network
- **Benefits:** Faster fine-tuning, lower computation and memory usage

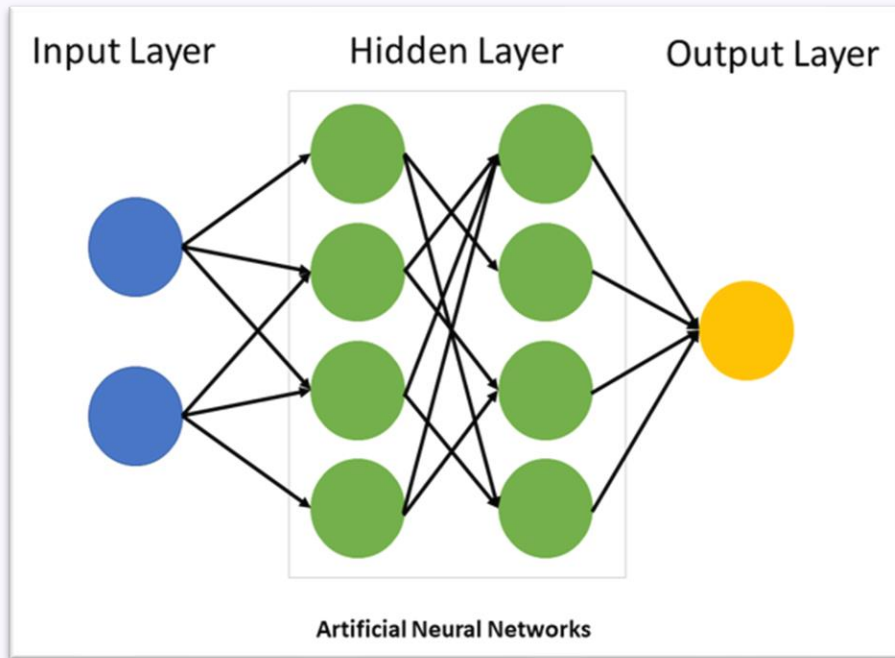


Image source: "Neural network explain" by TseKiChun, licensed under CC BY-SA 4.0. Available at: Wikimedia Commons.

Model Fine-Tuning

Supervised Training and Validation

- **Tool:** SFTTrainer for *supervised fine-tuning* of LLMs
- **Input:** Formatted dataset and prior configurations
- **Purpose:** Efficient training and validation
- **Use:** Employed for both training and validation with corresponding datasets

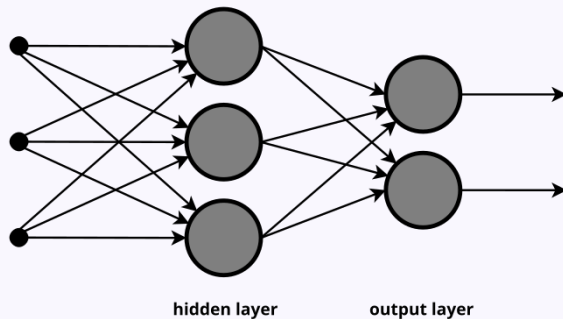


Image source: "Multi-Layer Neural Network-Vector.svg" available on Wikimedia Commons, public domain.

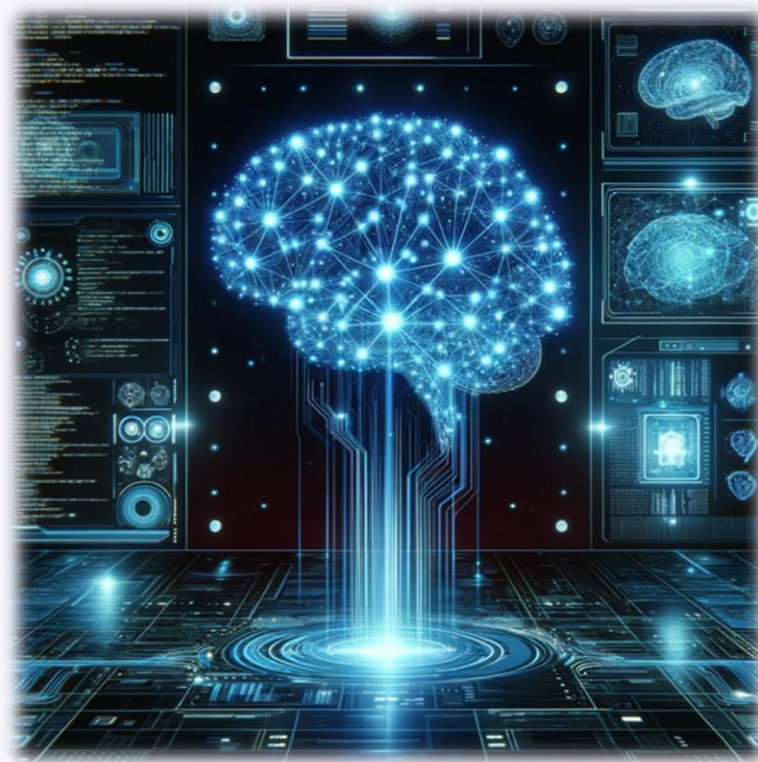
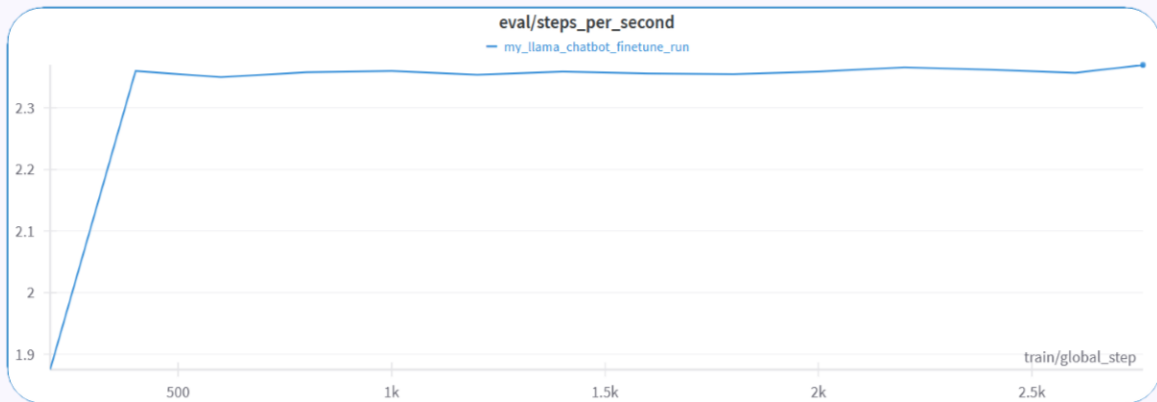


Image source: "DALL-E 3 - advanced artificial intelligence.png," generated by ChatGPT with DALL-E 3, released into the public domain by Alenoach

Model Fine-Tuning

Performance Evaluation and Checkpoints

- **Metrics:** *Loss* and *perplexity*
- **Evaluation:** Performed on validation dataset
- **Logging:** Every 200 *steps* for efficiency and monitoring
- **Checkpoints:** Models saved at 400, 1200, and 2000 based on loss values



Step	Training Loss	Validation Loss
200	0.196800	0.323974
400	0.121500	0.328974
600	0.091100	0.359895
800	0.074300	0.376075
1000	0.064700	0.410452
1200	0.060100	0.424658
1400	0.057800	0.409700
1600	0.053100	0.431398
1800	0.049700	0.458669
2000	0.048300	0.508953
2200	0.048000	0.532966
2400	0.047000	0.546369
2600	0.046500	0.556386



04

Chatbot Inference and Deploying

Chatbot Inference and Deploying

Hosting Platform

- **Initial setup:** *Google Colab* with free GPUs (15 GB RAM)
- **Transition:** Moved to *Hugging Face* for more resources
- **Free plan issue:** CPU only, slow responses (700 seconds/query)
- **Solution:** *Hugging Face Pro plan*, response time reduced to 15 seconds (mainly for model loading)



Image source: "Google Colaboratory SVG Logo.svg," original image by Google, released into the public domain. Retrieved from Wikimedia Commons (<https://commons.wikimedia.org>).



Hugging Face

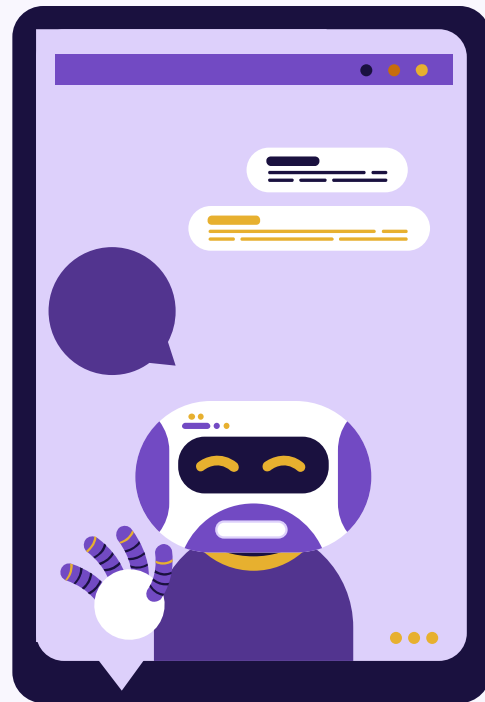
Image source: "Hf-logo-with-title.svg," Hugging Face logo by Victor (Hugging Face Staff), released into the public domain. Retrieved from Wikimedia Commons (<https://commons.wikimedia.org>).

Chatbot Inference and Deploying

Inference Mode

- **Model and tokenizer:** Saved on *Hugging Face Hub*
- **Inference:** Model answers questions about AROL Group
- **Focus:** Bottle caps, capping technologies, related services

PRE-THREADED FLAT PLASTIC CAPS  AROL's equipment is able to run typical beverage plastic caps from 1.500 bph up to 90.000 bph. MACHINES	STEEL CROWN CORK  Crowns are still widely used with glass, plastic, and aluminium bottles. AROL's traditional Crowner run both 26 and 29 crowns. MACHINES	ALUMINIUM CROWN CORK  AROL provides a crowner suitable for running the 26 and 29 aluminium crown cork up to 14.400 bph. MACHINES
--	--	--



Chatbot Inference and Deploying

Retrieval-Augmented Generation

- **Implementation:** RAG (Retrieval-Augmented Generation) for better performance
- **Storage:** *Pinecone* used for vector storage and search
- **Process:** Queries embedded as *vectors*, compared with stored vectors
- **Result:** Top 5 relevant documents retrieved
- **Generation:** LLM generates accurate responses using these documents
- **Pinecone:** Fast, efficient vector similarity searches



Image source: "Pinecone-Full-Logo-Black.svg," Logo of Pinecone, a software company, by Pinecone Inc, released under Creative Commons Zero, Public Domain Dedication. Retrieved from Wikimedia Commons (<https://commons.wikimedia.org>).

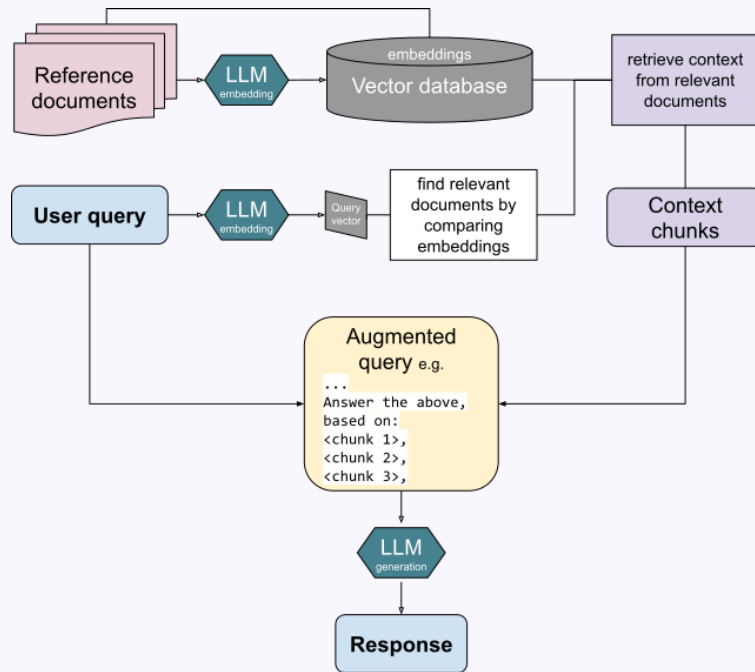


Image source: "RAG diagram.svg," Diagram: Retrieval-Augmented Generation (RAG) combines external documents and user input into an LLM prompt to get tailored output, by Turtlecrown, released under Creative Commons Attribution-Share Alike 4.0. Retrieved from Wikimedia Commons (<https://commons.wikimedia.org>).

Chatbot Inference and Deploying

Generation Pipeline

- **Chatbot pipeline:** Combined model with tokenizer
- **Output:** Clear, concise (150 tokens)
- **Context:** Enriched with external info
- **Tools:** RAG, Pinecone
- **Benefits:** Higher accuracy, domain-specific customization, flexible updates

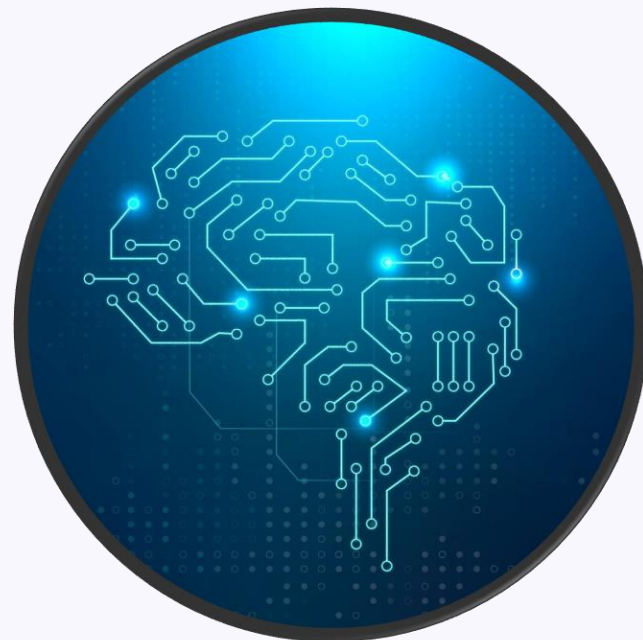
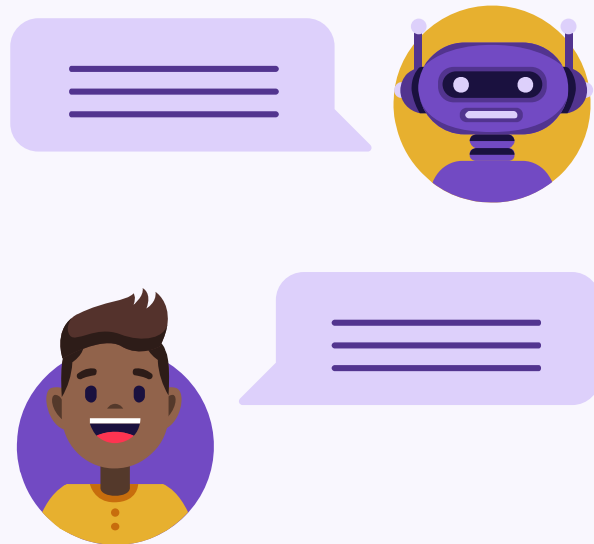


Image source: 'AI technology brain background vector digital transformation concept,' designed by Freepik. Attribution required as per the license. Retrieved from Freepik (<https://www.freepik.com>).

Chatbot Inference and Deploying

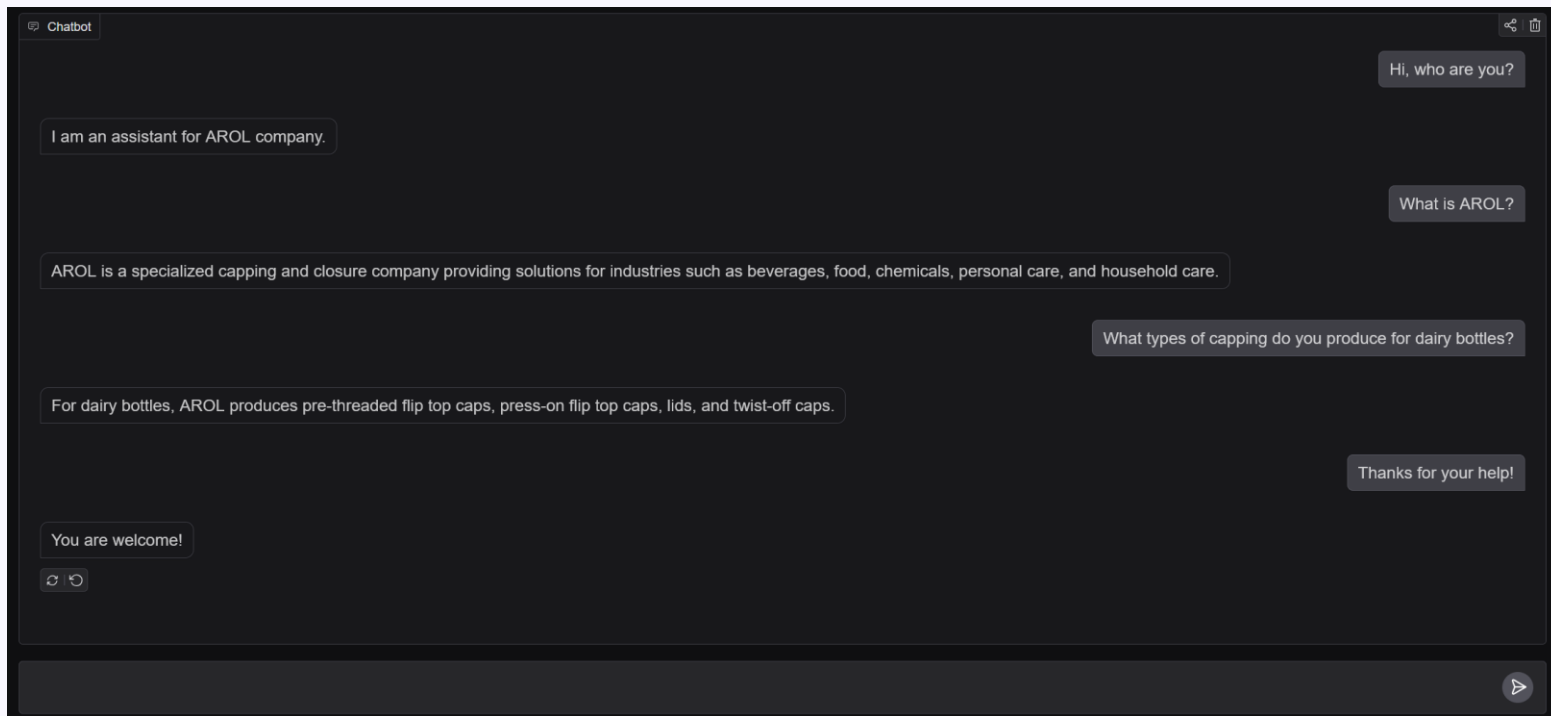
Deploying

- **Deployment:** *Hugging Face Spaces* with *Gradio*
- **Purpose:** Quick demo/web app for ML model
- **Interface:** Chat-based, user-friendly, browser-accessible
- **Features:** Real-time input/output, extended logging (track messages/errors)



Chatbot Inference and Deploying

Demo

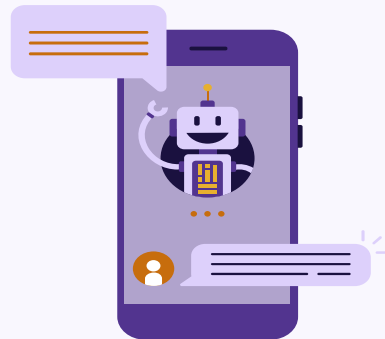


Try the demo at: <https://huggingface.co/spaces/Meldashti/unsloth-llama-3-8b-bnb-4bit>

Thanks!

Does anyone have any questions?

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For more information and to access all project materials, visit the following links:

- **Project Repository (Code & Documentation):** <https://github.com/MelDashti/Arol-ChatBot>
- **Fine-Tuned Model on Hugging Face Hub:** <https://huggingface.co/Meldashti/chatbot/tree/main>
- **Chatbot demo:** <https://huggingface.co/spaces/Meldashti/unsloth-llama-3-8b-bnb-4bit>

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